

Appendix E: how we priced the note, and why the fair coupon is 13.75%

Monte Carlo simulation of the five-ETF basket (1,000,000 paths); the fair coupon makes the note worth exactly its issue price

A	Term structure & discount factors Annual rates are converted to continuous rates; each cash flow is discounted at its own tenor	$r_j = \ln(1 + R_j) \quad (1)$ $D_j = \exp(-r_j \cdot t_j) \quad (2)$
B	Asset paths (correlated geometric Brownian motion) Each ETF moves one year at a time with its own interval variance ΔV ; shocks are correlated through the Cholesky factor L of ρ	$X_{i,j} = X_{i,j-1} \cdot \exp((r_j - q_{i,j} - \frac{1}{2}\Delta V_{i,j}) + \sqrt{\Delta V_{i,j}} \cdot Z_{i,j}) \quad (3)$ $Z = \varepsilon \cdot L^T, \quad \rho = L \cdot L^T \quad (4)$
C	Basket & autocall trigger The basket is the weighted sum of normalised ETF prices; the note ends at the first year it reaches that year's barrier	$B_j = \sum_{i=1..5} w_i X_{i,j} \quad (5)$ call at first k with $B_k \geq H_k \quad (6)$
D	Maturity payout (never called) A 70% barrier observed only at Year 5	$R_5 = 1.0 \text{ if } B_5 \geq 0.70; \quad R_5 = B_5 \text{ if } B_5 < 0.70 \quad (7)$
E	Fair coupon a = discounted principal repaid, b = discounted years of coupon (Case A: no coupon on never-called paths)	$1.0 = E[a + y \cdot b] \quad (8)$ $y = (1.0 - E[a]) / E[b] \quad (9)$

Product priced	
Basket weights	3119.HK 35% · EWY 20% · EWT 15% · 2644.T 15% · 00878.TW 15%
Autocall barrier H	Y1 102.5% · Y2 100% · Y3 97.5% · Y4 95% · Y5 92.5%
Called in year k	$100 \times (1 + k \times y)$
Never called	R_5 as in (7)
Simulation	1,000,000 paths, annual steps

Pricing results (Case A)	
Fair coupon	13.752219%
Standard error	2.236 bp
Approx. 95% CI	[13.708396%, 13.796041%]

13.752219% = 10% coupon to the client + 3.752219% kept by Natixis

We are collecting 3.752219% from the customer and give them a 10% coupon

Source: team write-up 'Derivation of Fair Coupon Rate for Autocallable Structured Product' (8 Oct 2026), sections 2–3. Excludes issuer credit, fees and hedging costs.

Appendix F: pricing inputs, grids and matrices

Every input used in the Monte Carlo simulation (team write-up, section 4)

4.1 Term structure of interest rates			
Year (t)	Risk-free rate (R)	Continuous rate $r = \ln(1 + R)$	Discount factor (D)
1	4.78%	4.671%	0.9544
2	5.30%	5.166%	0.9015
3	5.52%	5.375%	0.8466
4 (interpolated)	5.605%	5.454%	0.7981
5	5.69%	5.534%	0.7510

4.2 Correlation matrix (ρ)					
Asset	3119.HK	EWY	EWT	2644.T	00878.TW
3119.HK	1.000	0.534	0.741	0.863	0.562
EWY	0.534	1.000	0.587	0.491	0.438
EWT	0.741	0.587	1.000	0.614	0.689
2644.T	0.863	0.491	0.614	1.000	0.517
00878.TW	0.562	0.438	0.689	0.517	1.000

4.3 Implied volatility grid (σ), annualised					
Asset / Year	Year 1	Year 2	Year 3	Year 4	Year 5
3119.HK	30.45%	32.34%	33.50%	36.70%	37.50%
EWY	41.50%	42.70%	44.50%	44.50%	44.50%
EWT	32.40%	35.80%	36.30%	36.30%	36.30%
2644.T	23.00%	24.90%	25.60%	25.60%	25.90%
00878.TW	7.00%	9.00%	12.00%	12.00%	12.00%

4.4 Interval variances (ΔV), drive each year's diffusion step					
Asset / Interval	Year 1	Year 2	Year 3	Year 4	Year 5
3119.HK	0.0927	0.1126	0.1177	0.1633	0.1744
EWY	0.1722	0.1868	0.2078	0.2078	0.2078
EWT	0.1050	0.1340	0.1384	0.1384	0.1406
2644.T	0.0529	0.0620	0.0655	0.0655	0.0689
00878.TW	0.0049	0.0081	0.0162	0.0162	0.0162

Basket weights 3119.HK 35% · EWY 20% · EWT 15% · 2644.T 15% · 00878.TW 15% | Autocall barriers 102.5% / 100% / 97.5% / 95% / 92.5% | Maturity barrier 70%

Source: team write-up, section 4. Rates: case USD funding grid (Year 4 interpolated). Correlation matrix checked positive definite before the Cholesky decomposition.